
EE 66
Fall 2026

Signals, Dynamics, and Information

Discussion 3B

1. DTFS Expansion

Determine the complex-exponential discrete-time Fourier series (DTFS) expansion for the signal $x : \mathbb{Z} \rightarrow \mathbb{R}$ described below, or explain why no such expansion exists. For the case where a DTFS expansion exists, be sure to identify the period p and the fundamental frequency ω_0 .

(a) In this problem, we introduce the Kronecker-Delta function δ . This function is defined as follows:

$$\delta[n] = \begin{cases} 1 & \text{if } n = 0 \\ 0 & \text{if } n \neq 0 \end{cases}$$

Let $x[n] = \sum_{\ell=-\infty}^{\infty} \delta[n - 4\ell]$ be a signal, for all integers n . To start, provide a well-labeled plot of the signal x first, so you can also see its structure. Then, perform the DTFS expansion, if it exists.

(b) $x[n] = \cos\left(\frac{2\pi}{5}n\right), \quad \forall n.$

(c) $x[n] = \sin\left(\frac{2\pi}{5}n\right) + \cos\left(\frac{4\pi}{5}n\right), \quad \forall n.$

2. Orthogonality of Discrete-Time Complex Exponentials (Fourier Basis)

Let

$$x_k[n] = e^{i\frac{2\pi}{p}kn}, \quad n = 0, 1, \dots, p-1$$

for $k = 0, 1, \dots, p-1$. The complex inner product on $\mathbb{C}^p$ is

$$\langle a, b \rangle = \sum_{n=0}^{p-1} a[n] b^*[n].$$

(a) Show that

$$\langle x_k, x_m \rangle = 0 \quad \text{whenever } k \neq m.$$

(b) Explain why this orthogonality makes the set

$$\{x_k[n] = e^{i\frac{2\pi}{p}kn} \mid k \in [0, p-1]\}$$

a natural *basis* (the discrete Fourier basis) for $\mathbb{C}^p$.